Name	**Gayathri. P**
Reg no	**192424324**
Course	**Mobile Computing**
Code	**ITA0329**

ASSIGNMENT 2 – CO2

Q1 – Mobile Communication or Mobile Computing?

Question

A utility company uses (a) a rugged tablet with cellular data, (b) a laptop connected to Ethernet at different offices, (c) a fixed wireless router, (d) a push-to-talk handset and (e) an offline meter-reading tablet that synchronises later. Explain the difference between wireless and mobile, classify the devices, give one ambiguous example and explain the offline-sync test.

Answer:

Mobile communication means transferring voice or data through a communication network while using a mobile device. Mobile computing means using a portable device to collect, process, store and access information while moving or even when the network is unavailable. Wireless means that a connection uses radio signals instead of a cable. Therefore, wireless and mobile do not have the same meaning.

Classification criteria

- Mobility: whether the user or device can move while being used.

- Connection type: whether the connection is wired or wireless.
- Disconnection tolerance: whether the application can work without a network.
- Local storage and processing: whether the device can save and process information locally.
- Location awareness: whether the application uses the current location or context.

Case	**Classification**	**Reason**
(a) Cellular rugged tablet	Mobile	The worker can move and the tablet receives work orders through cellular data.
(b) Laptop with Ethernet at different offices	Nomadic	The laptop is moved between offices, but it uses a wired connection while working.
(c) Fixed wireless router	Fixed wireless	It uses a wireless connection, but the router remains at a fixed location.
(d) Push-to-talk handset	Mobile	The user can move while communicating by voice.
(e) Offline meter-reading tablet	Mobile	It stores readings locally and synchronises them when a connection becomes available.

Additional ambiguous case

A delivery driver uses a tablet fixed inside a moving vehicle. It receives delivery updates through cellular data and uses GPS for navigation. It is mobile because the vehicle moves, and it supports both mobile communication and mobile computing.

Offline synchronisation test – explanation

The assignment asks to compare immediate sending and store-and-forward at 90%, 60% and 30% link availability. In immediate sending, an update is delivered only when the network is available; otherwise it is lost. In store-and-forward, an update is saved in a queue and sent when the network returns. Therefore, store-and-forward is expected to deliver more updates, but it may increase delivery delay. As availability decreases, immediate-send losses and store-and-forward delays are expected to increase.

Conclusion: Mobile computing needs local storage, retry mechanisms, data synchronisation and duplicate checking. A wireless connection alone cannot provide these features.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic

Question

Compare FDMA and TDMA for a 2 MHz band. Explain their usable capacity, performance for continuous voice, bursty data and mixed traffic, and decide whether dedicated channels are suitable for low-rate campus sensors.

Answer:

FDMA (Frequency Division Multiple Access) divides the available frequency band into separate frequency channels. Each user gets a different frequency channel. TDMA (Time Division Multiple Access) divides a carrier into time slots, and different users transmit in different slots.

Nominal capacity calculation

- Total bandwidth = 2 MHz = 2,000 kHz.
- FDMA channels = 2,000 / 25 = 80 nominal channels.
- TDMA carriers = 2,000 / 200 = 10 carriers.
- Each TDMA carrier has 8 slots. Therefore, 10 × 8 = 80 nominal time slots.

Both methods have 80 nominal channels or slots. However, the usable capacity can be lower because of guard bands, guard times, synchronisation and control information. The exact usable capacity cannot be calculated from the question alone because it does not specify the overhead values. These values must be stated as assumptions or taken from a cited standard.

Traffic type	**Expected suitable method**	**Reason**
Continuous voice	FDMA or TDMA	A continuous call can use its assigned resource regularly. Both can support voice.
Bursty data with 10% duty cycle	Dynamic TDMA	Time slots can be assigned when data is ready. A permanently assigned frequency may remain idle.
70% voice and 30% data	TDMA can be flexible	Regular slots can support voice while other slots can be used for data bursts.

Three traffic conditions

- Continuous voice: calls remain active for a long period.

- Bursty telemetry: devices send short reports occasionally, with long idle periods.
- Synchronised arrivals: many devices send data at the same time, creating a sudden traffic peak.

In continuous voice, dedicated channels can be used efficiently. In bursty traffic, dedicated channels may remain idle. In synchronised arrivals, both methods may experience blocking if the number of requests is greater than the available resources.

Campus telemetry decision

Dedicated FDMA or TDMA channels are not justified when 90% of devices send only a 200-byte report per minute. Most channels would remain unused between reports. A shared or dynamically scheduled access method is more suitable. A small amount of capacity can be reserved for important reports.

Conclusion: Equal nominal capacity does not mean equal usable capacity. Traffic pattern and overhead decide the actual efficiency.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem

Question

Explain a direct-sequence CDMA uplink, spreading and despreading, processing gain, orthogonal and random codes, the near-far problem and the importance of power control.

Answer:

CDMA stands for Code Division Multiple Access. It allows multiple users to share the same frequency band and time by assigning a different spreading code to each user. In direct-sequence CDMA, each data bit is multiplied by a sequence of smaller units called chips.

Spreading and despreading

- At the transmitter, the user's data bit is multiplied by its spreading code.
- The data is spread over a larger number of chips.
- At the receiver, the received signal is multiplied by the same user's code.
- The receiver adds the chip values and decides whether the original bit was positive or negative.

Processing gain

Processing gain is the ratio of chip rate to data rate. In the simple case where one data bit is spread using N chips, processing gain is N. Therefore, code lengths 8, 16 and 64 provide processing gains of 8, 16 and 64 respectively.

Expected behaviour under different conditions

Condition	**Expected behaviour**
Synchronous orthogonal Walsh codes	Different users ideally have zero cross-correlation, so interference is low.
Random non-orthogonal codes	Codes may have non-zero correlation. Interference and bit errors can increase as the number of users increases.
One user has ten times the power	The strong user's signal can affect detection of a weak user's signal. This is the near-far problem.

Inquiry answers

- Orthogonal codes may lose their orthogonality when users are not synchronised because their chip timings do not match.
- Unequal received power causes a strong signal to dominate a weak signal at the base station.

- Power control is important because it tries to make the received signal powers of different users nearly equal.
- The number of supported users depends on code length, noise, synchronisation, power control and the required bit-error rate. It cannot be stated accurately without experimental results.

Conclusion: CDMA allows users to share the same channel using codes. However, synchronisation and power control are necessary to reduce interference.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best

Question

Explain subcarrier spacing, symbol duration, cyclic prefix and orthogonality. Compare fixed contiguous, interleaved round-robin and channel-aware allocation. Explain throughput and fairness.

Answer:

OFDMA stands for Orthogonal Frequency Division Multiple Access. It divides a wide channel into many small subcarriers. Different users can be assigned different groups of subcarriers, allowing them to communicate at the same time.

Important concepts

- Subcarrier spacing and useful symbol duration are related by $\Delta f = 1/Tu$.
- A cyclic prefix is a copy of the end part of an OFDM symbol added to its beginning.
- The cyclic prefix helps reduce inter-symbol interference caused by multipath propagation.
- Orthogonality means that different ideal subcarriers do not interfere with one another over the useful symbol duration.

Three allocation methods

Method	Explanation	Expected result
Fixed contiguous blocks	Each user receives a continuous group of nearby subcarriers.	Simple allocation, but a user may receive subcarriers with poor channel quality.
Interleaved round-robin	Subcarriers are distributed among users in a repeating order.	Provides frequency diversity and can improve fairness.
Channel-aware best-subcarrier	Each subcarrier is assigned to the user with the best channel quality on it.	Can increase total throughput, but users with weak channels may receive fewer resources.

Throughput and fairness

Throughput is the amount of data successfully transmitted in a given time. Channel-aware allocation may provide higher total throughput because it assigns subcarriers to users with better channel conditions. However, it may not distribute resources equally.

Jain's fairness index is used to measure fairness. Its value is between 0 and 1. A value closer to 1 means that throughput is shared more equally among users.

Design decision

For a lecture hall, I would select a fairness-aware allocation. Every student should receive a minimum guaranteed data rate. The remaining resources can be allocated based on channel quality. This avoids giving all the resources only to users with strong signals.

Conclusion: OFDMA improves resource sharing, but the system must balance total throughput and fairness.

Q5 – SDMA – Reusing the Same Channel in Different Directions

Question

Explain SDMA using omni antennas, sector antennas and a uniform linear antenna array. Discuss array factor, beamwidth, side lobes, SIR and the limitations of SDMA.

Answer:

SDMA stands for Space Division Multiple Access. It allows multiple users to use the same frequency and time resource by separating them according to their spatial directions. Directional antennas or antenna arrays are used to focus signals toward particular users.

Antenna types

- Omni antenna: radiates or receives signals in all horizontal directions. It provides broad coverage but has limited spatial separation.
- 120-degree sector antenna: covers a particular sector and reduces interference from other directions.
- Uniform linear array (ULA): uses multiple antenna elements in a line. The beam can be steered toward a selected user.

Array factor and beamwidth

The array factor describes how the signals from the antenna elements combine in different directions. The main lobe is the strongest beam directed toward the intended user. Side lobes are smaller unwanted beams in other directions.

When the number of antenna elements increases, the main beam generally becomes narrower and directional separation improves. The side-lobe level depends on the antenna spacing and element weighting.

SIR and angular separation

SIR means Signal-to-Interference Ratio. It compares the desired signal power with interference power. If two users are separated by a sufficient angle, the antenna can direct a strong beam toward one user and reduce the signal received from the other. The assignment suggests testing separations from 5° to 60° and selecting an SIR threshold. The exact minimum separation depends on the array size, antenna pattern, power and selected threshold.

Limitations and toll-plaza decision

- SDMA is less effective when users are located in nearly the same direction.
- Angular spread and groups of closely located users make separation difficult.
- At a toll plaza, most vehicles arrive from a similar direction. Therefore, I would not depend on SDMA alone.

I would combine SDMA with time or frequency scheduling to separate vehicles that cannot be distinguished by direction.

Conclusion: SDMA improves channel reuse when users are spatially separated. It becomes difficult when users are clustered in the same direction.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

Question

Select suitable access methods for cellular voice, a jam-resistant tactical link, a high-rate stadium downlink and 5,000 low-rate sensors. Define weighted criteria and explain sensitivity analysis.

Answer

Different applications have different requirements. Therefore, no single multiple-access technique is best for every situation. I use five criteria to compare the techniques.

Criterion	Weight	Reason
Capacity / spectral efficiency	25%	Important when many users share limited bandwidth.
Delay support	20%	Important for voice and real-time communication.
Interference resistance	20%	Important in crowded or disturbed radio environments.
Scalability	20%	Important when the number of devices increases.
Implementation complexity	15%	Important for cost, maintenance and deployment.

A rating can be given from 1 to 5. A score of 1 means poor suitability and 5 means excellent suitability. The weighted score is calculated by multiplying each rating by its criterion weight and adding the results.

Suggested deployment choices

Deployment	Suitable technique	Reason
Wide-area cellular voice	OFDMA with SDMA support	OFDMA shares frequency resources efficiently, while SDMA can improve spatial reuse.
Jam-resistant tactical link	CDMA, with suitable additional protection	Spreading can help resist some narrowband interference. No access method alone guarantees protection against jamming.
High-rate stadium downlink	OFDMA with SDMA/MIMO	It can support many users and provide high data rates through frequency and spatial resource allocation.
5,000 low-rate sensors	Scheduled or shared packet access	Small, occasional reports do not need a permanently dedicated channel for every sensor.

Sensitivity analysis

Sensitivity analysis checks whether the selected technique changes when the weights are increased or decreased by 20%. If the same technique remains the highest-scoring choice, the decision is stable. If another technique becomes the winner, the decision is sensitive to the changed criterion.

Real systems combine techniques because each technique has different strengths. For example, FDMA can divide a band into carriers and TDMA can divide each carrier into time slots. A hybrid design can use resources more effectively than relying on only one method.

Conclusion: The best technique depends on capacity, delay, interference, number of users and implementation cost. The final selection should be supported by the comparison and sensitivity analysis.

Q7 – Diagnose, Redesign and Validate

Question

Choose one limitation in mobile communication technology. Diagnose the problem, redesign the communication strategy, predict the improvement, validate the redesigned approach, and justify your design.

Chosen problem: Dedicated channels are wasted under bursty traffic

I selected the limitation of dedicated FDMA/TDMA channels under bursty traffic. In this design, a device keeps its assigned channel even when it has no data to send. As a result, the channel remains idle and available capacity is wasted.

A. Diagnose the bottleneck

- The main problem is the reservation of channels for devices that are not transmitting.
- The amount of idle reserved channel-time increases when many devices send data only occasionally.
- Resources are wasted during the waiting period between reports.
- The fixed allocation method does not adjust to changing traffic.

B. Define the design objective

The objective is to reduce idle channel usage and deliver sensor reports within an acceptable delay. The redesigned system should allocate a channel only when a device has data to send and release it after transmission.

C. Revised strategy

I propose a shared on-demand channel allocation method. All devices place their packets in a queue. The scheduler checks for free channels and assigns them to devices with pending packets. After transmission, the channel is released and can be used by another device.

Steps:

1. Collect packets generated by the devices.
2. Place pending packets in a queue.
3. Check whether a channel is free.
4. Assign a free channel to the oldest pending packet.
5. Transmit the packet and release the channel.
6. Repeat until all available channels are occupied or the queue becomes empty.

This method reduces unnecessary channel reservation. However, it requires queue memory and scheduling control. Some packets may experience a short waiting delay during heavy traffic.

D. Prediction before implementation

I predict that the redesigned method will reduce idle channel-time and improve resource utilisation for low-rate, bursty sensor traffic. During a sudden large burst, some packets may have to wait.

E. Validation

The original and redesigned methods should be compared using the same traffic pattern and the same number of available channels. The following measures can be compared:

- Number of packets generated.
- Number of packets delivered.
- Number of packets blocked or still waiting.
- Channel-time used and idle channel-time.
- Packet waiting time and deadline misses.

The assignment requires actual experimental evidence for these values. Since this written version does not include implementation, the measured values should not be invented.

F. Final design defence

The redesigned method is expected to be more efficient when devices send small reports occasionally because channels are assigned only when needed. Its disadvantages are additional scheduling complexity and possible waiting delay. Fixed allocation may still be better for continuous voice or services that need a guaranteed resource.

Conclusion: On-demand allocation is suitable for bursty sensor traffic because it reduces the waste caused by permanently reserved channels. The final decision must be supported by measured results if experimental validation is required.